Formatting Guide for Slide Decks

Quarto + reveal.js conventions used in this repo

2026-11-02

This guide records the formatting conventions used in the research-talk decks in this repo (e.g., `usf.qmd`). Examples are shown as literal markdown so you can copy them into a deck.

Workflow and files

File	Role
`_quarto_uw.yml`	Shared reveal.js settings (CSS files, title-slide background, `auto-stretch: false`, `scrollable: true`). Copied to `_quarto.yml` only while publishing.
`presentation.css`	**Theme**: the automatic look of every slide (fonts, header colors, red right border, list bullets, title slide).
`utilities.css`	**Utilities**: opt-in classes you use in the `.qmd` (colors, spacers, table highlight, callout placement). Intended to stay identical to the copy in `~/github/book_git` (see `HANDOFF.md`).
`publish.sh`	Publishes one deck to Posit Connect Cloud.
`_publish.yml`	Written by `quarto publish`; maps each deck to its Connect Cloud item. Do not hand-edit.
`figs/`, `mak/`, `data/`	Figure scripts, data-prep scripts, and locally cached data (see Figures from R).
`images/`	Static images (photos, logos, screenshots).

There is **no committed `_quarto.yml`**. Publish with:

```
./publish.sh usf.qmd
```

which copies `_quarto_uw.yml` to `_quarto.yml`, publishes, then deletes `_quarto.yml`, `*_files/`, and `*.html`. The first publish of a new deck creates a new Connect Cloud item and adds it to `_publish.yml`; later publishes update that item.

To preview locally without publishing:

```
cp _quarto_uw.yml _quarto.yml
quarto preview usf.qmd
rm _quarto.yml
```

Deck YAML header

Each deck sets only its own metadata; formatting comes from `_quarto_uw.yml`.

```
---
title: 'A "Smart" Recovery Monitoring and Support System for SUD Continuing Care'
author: "John J. Curtin, Ph.D."
institute: "University of Wisconsin-Madison"
date: "November 6, 2026"
editor_options:
  chunk_output_type: console
---
```

R code is hidden by default in reveal.js. Teaching decks that show code add `execute: echo: true` to the header.

Slides and headings

Slide breaks

Separate slides with a long horizontal rule. It makes slide boundaries easy to see in the source.

```
---------------------------------------------------------------------------


## Slide Title
```

Heading levels

Markdown	Renders as	Use
`#`	60px black, left aligned, red underline bar	Section title slides (rare in research talks)
`##`	40px UW red with a short red bar underneath	**Every slide title**
`###`	40px black	Rare
`#### Text {.uwred style="margin-top: 3rem;"}`	Red sub-heading with space above	A second heading lower on a slide (e.g., below a table)

Repeating the same `##` title over several slides is fine and is how progressive builds work (see Progressive builds).

Intentionally blank slides

A slide with no heading and no content gives a blank screen so the audience focuses on the speaker. Put what you plan to say in the notes.

```
-------------------------------------------------------------------------------

::: {.notes}
What I will say while the screen is blank.
:::

-------------------------------------------------------------------------------
```

A slide that has a title but no body (e.g., `## Next Steps: Algorithmic Fairness` followed only by notes) works the same way while keeping the section label on screen.

Full-slide background images

For an image that fills the slide with no title, put the attributes on an empty `##` heading. Give it an `id` and a `data-menu-title` so it is identifiable in the slide menu.

```
## {#gps_detection_1 data-menu-title="GPS detection, wide view" background-image="https://di
```

Title slide

The title slide is built from the YAML header. Its background image (`images/title_slide.png`) and white, left-aligned text come from `_quarto_uw.yml` and `presentation.css`; nothing is needed in the deck.

Speaker notes

Every slide should have notes. Press `s` during the presentation to open speaker view.

```
::: {.notes}
Notes text. Can include lists, **bold**, and [PAUSE] cues.
:::
```

Text and emphasis

The theme makes all body text bold at 25px; `**strong**` is even heavier (weight 900).

Color classes (`utilities.css`)

Class	Color	Typical use
`.uwred`	UW red `#c5050c`	Key terms in teaching decks
`.blue`	blue	Key terms in research talks
`.gray`	gray	Points already covered, in progressive builds
`.red`, `.green`, `.teal`	red, forestgreen, teal	Contrasts (e.g., correct vs. error)

Apply them to inline text with Pandoc spans, or with raw HTML inside HTML blocks:

```
Provide the [right interventions]{.blue} to the [right people]{.blue}

<span class="green">Correct (True Positive, TP)</span>
```

Use classes, not inline `style="color: ..."`.

Progressive builds

We build up content by **duplicating the slide** and changing the emphasis, not with reveal fragments. Typically, text already discussed turns gray and new text is added below.

```
## Personal Sensing

Personal sensing collects information from smartphones ...

---------------------------------------------------------------------------

## Personal Sensing

[Personal sensing collects information from smartphones ...]{.gray}

\

We focus on

- Ecological Momentary Assessment (EMA)
- Geolocation
```

Graying a list item: wrap the item text, e.g., `- [Can be sustained for long periods]{.gray}`.

Block quotes

```
> "Could you predict not only [who]{.blue} might be at greatest risk ... <br>
 ... but precisely [when]{.blue} that relapse might occur ..."
```

Spacing

From coarsest to finest:

Method	Markdown	Effect
Blank paragraph	a line containing only \	One line of vertical space; stack two for more

Method	Markdown	Effect
Spacer div	`<div class="gap-xs"></div>` (also `gap-sm`, `gap-md`, `gap-lg`)	0.25, 0.5, 1, or 1.5 rem of vertical space
Line break in text	` `	Breaks a line within a paragraph or bullet, controlling where long lines wrap

```
- First point

\

- Second point, separated by one blank line

<div class="gap-sm"></div>

- Third point, separated by a smaller gap
```

Use \ lines in research talks and gap divs when you need finer control. Put a blank line before and after each.

Columns

```
::: {.columns}
:::: {.column width="40%"}
Left content
::::

:::: {.column width="60%"}
Right content (often a figure)
::::
:::
```

Use more colons on the inner fences so the nesting is easy to read. An empty column can hold a placeholder comment, `<!-- blank -->`, to reserve space for content that appears on a later slide of a build.

Callouts

Callouts render normally (in the page flow) by default. Wrap them to change placement.

Corner callout

`.callout-corner` pins a small (300px) box to the bottom-left of the slide. We use it for reference information that stays put across a build (e.g., auROC rules of thumb).

```
:::::: {.callout-corner}
:::::{.callout-tip icon=false}
## Rules of thumb for auROC
.70 - .80 is considered fair
.80 - .90 is considered good
\>= .90 is considered excellent
:::::
::::::
```

Inside columns, add the class to the column instead of adding another wrapper:

```
:::: {.column width="40%" .callout-corner}
:::::{.callout-tip icon=false}
## Rules of thumb for auROC
...
:::::
::::
```

Full-width callout

`.callout-wide` forces a full-width box in the normal flow. We use it for questions and interactive moments.

```
::: {.callout-wide}
::: {.callout-important}
# Question
Should you trust a significant effect less if power was low?
:::
:::
```

Callout details

- The first heading inside the callout becomes its title.
- Callout titles and text are shrunk to 75% by `presentation.css`.
- `icon=false` removes the icon.
- End lines with two spaces for line breaks inside a callout without bullets (as in the rules of thumb above).
- Types: `.callout-note`, `.callout-tip`, `.callout-important`, `.callout-warning`, `.callout-caution`.

Figures from R

Figures are made in separate scripts and only *displayed* in the deck, which keeps the `.qmd` readable and lets several decks reuse a figure.

Three-layer structure

Folder	Script	Job
`mak/<project>/`	`mak_*.R`	Pull results from the research drive (via `format_path()`) and cache them in `data/<project>/`. Each script checks `if (!file.exists(...))` first, so it only runs once per clone.
`data/<project>/`	(cached `.csv` / `.rds`)	Local copies of the results. Git-ignored; recreated by the `mak_*` scripts in each clone.
`figs/<project>/`	`figs_*.R`	Source the needed `mak_*` script, read from `data/`, and create named ggplot objects (e.g., `fig_auroc_ci_1`). They do not print anything.

Projects so far: `risk` (AUD/EMA study), `risk2` (NIDA OUD study), `opt` (optimization trial), `gen`.

`format_path()` comes from the lab_support repo and turns a study-drive path into the right mount point for Windows, Mac, or Linux. Load it with base R:

```r
source("https://github.com/jjcurtin/lab_support/blob/main/format_path.R?raw=true")
path_models <- format_path("risk2/models/combined")
```

Rendering therefore needs a network connection and, the first time in each clone, access to the research drive.

Displaying a figure

Source the script in the first chunk that needs it; the objects remain available to later chunks in the same deck. Use `here::here()` for paths.

````
```{r}
#| label: figs_auroc_ci_1
#| fig-height: 6
#| fig-width: 6

source(here::here("figs/risk/figs_ema_probability.R"))
fig_auroc_ci_1
```
````

- Give every chunk a unique `label` (labels must not repeat within a deck).
- Size with `fig-height` and `fig-width` (inches). These set the figure's size and aspect ratio, and how large plot text looks relative to it. `auto-stretch: false` is set, so reveal.js does not stretch figures to fill leftover space.
- Common sizes: full slide `10 x 6`; figure in a half or 60% column `6 x 6`; small figure in a 2 x 2 grid `5 x 3`.
- Combine several plots in one figure with `cowplot::plot_grid()`:

````
```{r}
#| label: figs_fairema_ci-2
#| fig-height: 6
#| fig-width: 10

cowplot::plot_grid(fig_fair_race_1, fig_fair_income_1, fig_fair_sex_1, nrow = 1)
```
````

- Figure captions are hidden by the theme, so do not rely on `fig-cap`.
- Style plots in the figure script (e.g., titles in `#C5050C`, larger axis text) so they match the deck.

Images

Inline images

Point at the image on GitHub with `?raw=true`, so the deck works wherever it is rendered:

```
![](https://github.com/jjcurtin/slide_decks/blob/main/images/ema_hand.png?raw=true){.absolute
```

A relative local path also works (e.g., `![](images/ema_hand.png)`); `quarto publish` uploads it with the deck. The image must be committed and pushed before a GitHub URL will work.

Positioning with `.absolute`

`.absolute` takes the image out of the text flow and places it relative to the slide. Combine it with any of `top`, `bottom`, `left`, `right` (pixels unless units are given) and `width` / `height`.

Goal	Attributes
Photo in the bottom-right corner	`{.absolute bottom=0 right=50 width=30%}`
Full-width banner along the bottom (e.g., PI/team strip)	`{.absolute bottom=0}`

Absolutely positioned images can overlap text, so check each slide after rendering.

Footers

A per-slide footer (right-aligned, bottom) for links, citations, or asides:

```
::: {.footer}
https://jjcurtin-power.share.connect.posit.cloud/
:::
```

Tables

Hand-written HTML tables let you highlight rows or cells, which is useful for builds:

```
<table>
<tr><th>Model</th><th>auROC</th></tr>
<tr class="hl"><td>Full</td><td>0.94</td></tr>
<tr><td>No geolocation</td><td>0.94</td></tr>
</table>
```

`class="hl"` on a `<tr>` fills the row with yellow (`#ffe08a`); on individual `<td>`/`<th>` cells it highlights a column.

Code output

When decks show code (`echo: true`), printed output is shown on a honeydew background by the theme. No markup is needed.

Gotchas

- **Close every HTML comment.** `<!--` without `-->` silently hides everything up to the next `-->`, including `:::` fences, which breaks the layout of the following slides.
- **Balance the colons.** Each opening fence needs a matching closing fence with the same number of colons.
- **Untitled slides are blank by design.** Do not "fix" them.
- `scrollable: true` is set, so overfull slides scroll instead of being cut off. If a slide scrolls during practice, trim it.
- **A leftover `_quarto.yml`** (e.g., from an interrupted publish) changes how every file in the folder renders. Delete it if a publish fails partway.
- **`publish.sh` deletes all `*.html` and `*_files/`** in the folder, including other rendered decks.